

ADVANCED STRENGTH OF MATERIALS

Time Allowed: 2 Hours

Full Marks: 35

Answer to Question No.1 is compulsory and to be answered first.**This answer is to be made in separate loose script(s) provided for the purpose.****Maximum time allowed is 30 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided.****On early submission of answer scripts of Question No.1,
a student will get the remaining script earlier.****Answer any five questions from the rest.**

1. Answer the following questions (any ten):

1×10=10

Fill in the blanks:

- i) Bending stress at N.A. is _____.
- ii) The section modulus of a beam having width b and depth d is given by _____.
- iii) The ratio of the polar moment of inertia to the radius of the shaft is called _____.
- iv) Force required to deflect a spring through unit deflection is called _____.
- v) In a strained body, maximum number of principal planes may be _____.
- vi) The maximum amount of strain energy that can be stored in a body without deformation is called _____.
- vii) A shaft is said to be in pure in pure tension, if it is subjected to _____ only.

State whether the following statements are True or False:

- viii) The stiffness of spring is also called spring constant.
- ix) The product of $G \times I_p$ is called torsional rigidity.
- x) The neutral axis of a beam is subjected to maximum tensile stress.
- xi) Core or kernel is the area within which the resultant of the load may be applied to avoid tensile stress.
- xii) Polar moment of inertia of a circular section of diameter D is $\pi/32 D^4$
- xiii) If section modulus of a beam is increase, the bending stress will increase.
- xiv) For a beam with rectangular section, maximum shear stress occurs at neutral axis.
- xv) If a coil spring is cut into two halves, deflection of the spring will be half.

- 2. What is simple bending? What are the assumptions made in the simple bending? 2+3
- 3. A steel plate 12 mm thick is bent around a drum. If the maximum stress induced in the plate due to bending be 400 N/mm^2 , find the radius of the drum. Take $E = 2 \times 10^5 \text{ N/mm}^2$. 5
- 4. Strain energy of two elastic bars subjected to gradual loading having cross-sectionals area are 400 mm^2 and 800 mm^2 are same. Establish a relation between stresses of two bars. 5
- 5. An I-section beam of flanges $20 \text{ cm} \times 2 \text{ cm}$ and web $30 \text{ cm} \times 1 \text{ cm}$ is acted on by a shear force of 150 kN determine: i).the maximum and minimum shearing stress in the web. ii).the maximum shear stress in the flange. 5
- 6. The principal stress at a point in the section of a boiler shell are 80 N/mm^2 and 40 N/mm^2 . Find the normal, tangential and resultant stress across a plane through the point inclined at 50° to the plane carrying 80 N/mm^2 stress. 5

7. Deduce torsion equation i.e. $\frac{T}{I_p} = \frac{\sigma_s}{R} = \frac{G\theta}{L}$ for solid shaft. 5
8. A solid shaft of 80 mm diameter is to be replaced by a hollow shaft of external diameter of 100 mm. 5 Determine the internal diameter of hollow shaft if the same power is to be transmitted by the both shafts at the same angular velocity and shear stress. 5
9. A closed coiled helical spring having the mean coil diameter is 12 times than the wire diameter. The spring is to absorb 30000 N cm of energy with an extension of 15 cm. The maximum shear stress set up in the spring is not to exceed 10000 N/cm². Determine – (a) mean coil diameter, (b) number of turns. 5
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